

Course Name

Course Code: PH113
Physics

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Module-5

Introduction to Plasma

Plasma

Plasma exists in many forms in nature and has a widespread use in science and technology. It is a special kind of ionized gas and in general consists of: – positively charged ions (‘positive ions’), – electrons, and – neutrals (atoms, molecules, radicals). We call an ionized gas ‘plasma’ if it is quasi-neutral and its properties are dominated by electric and/or magnetic forces.

Plasma

Plasma is called the *fourth state of matter* after solid, liquid, and gas. It is a state of matter in which an ionized substance becomes highly electrically conductive to the point that long-range electric and magnetic fields dominate its behaviour.

Plasma is typically an electrically quasi-neutral medium of unbound positive and negative particles (i.e. the overall charge of a plasma is roughly zero).

Plasma

Although these particles are unbound, they are not "free" in the sense of not experiencing forces. Moving charged particles generate electric currents, and any movement of a charged plasma particle affects and is affected by the fields created by the other charges. In turn this governs collective behaviour with many degrees of variation. Plasma is distinct from the other states of matter.

Plasma

Two basic necessary (but not sufficient) properties of the plasma are:

- a) presence of freely moving charged particles, and
- b) large number of these particles.

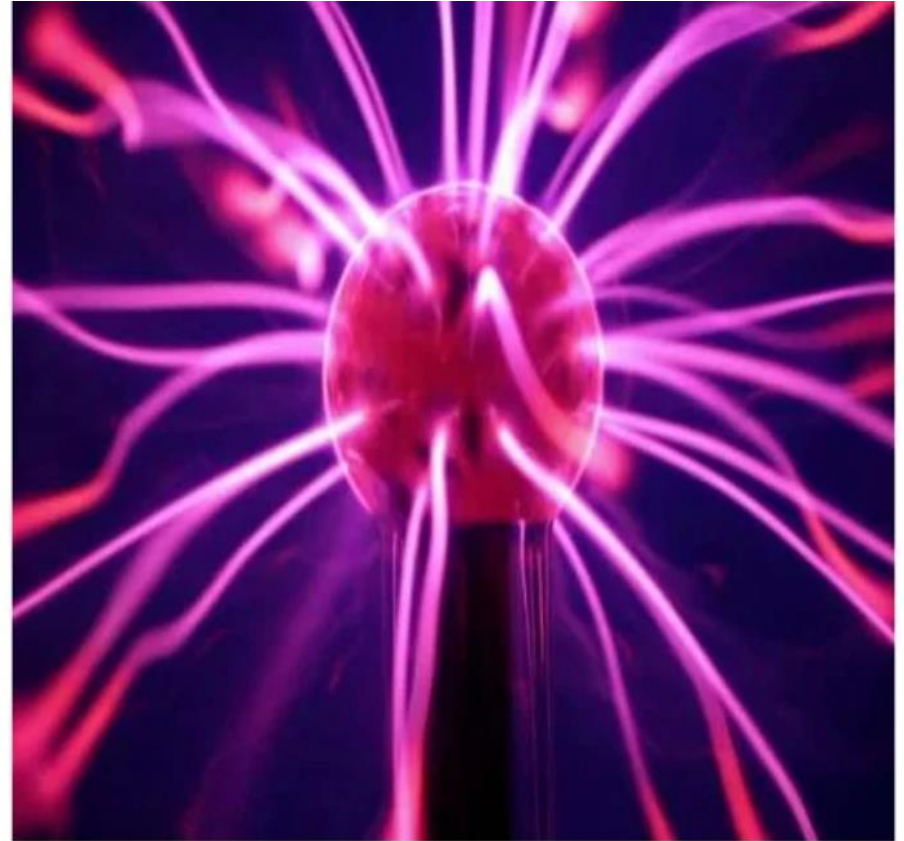
Plasma does not have to consists of charged particles only, neutrals may be present as well, and their relative number would affect the features of the system.

States of Matter: Plasma

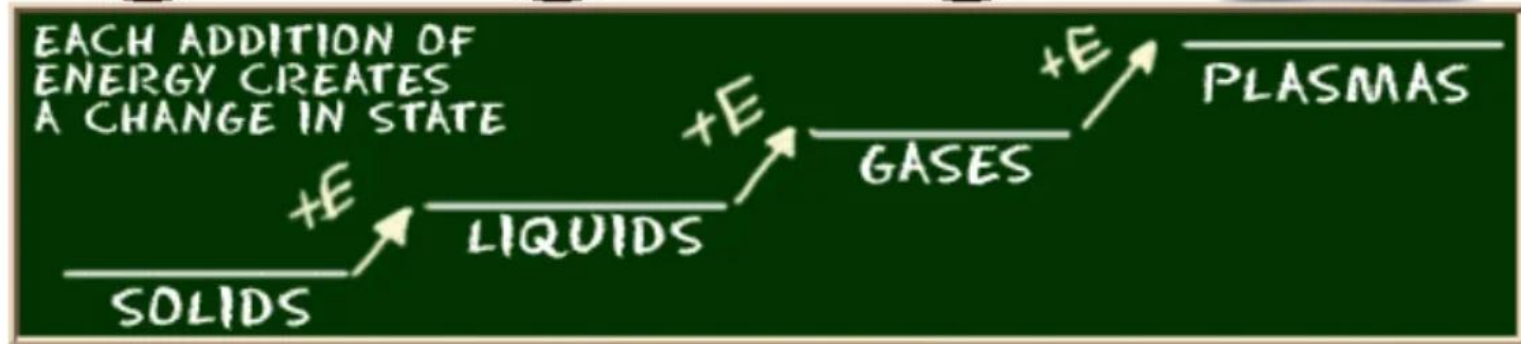
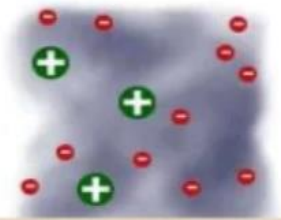
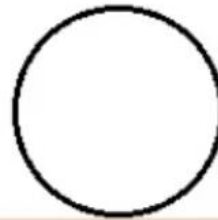
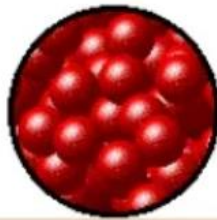
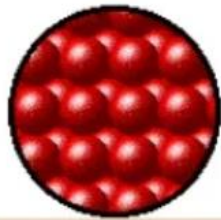
- A plasma is an ionized gas.
- A plasma is a very good conductor of electricity and is affected by magnetic fields.
- Plasma, like gases have an indefinite shape and an indefinite volume.

What is Plasma

- Plasma is considered 4th State of Matter despite solids, liquids and gases. It is one of the fundamental states of matter. Technically, it is an ionized gas consisting of positive ions and free electrons ,typically at low pressures (as in the upper atmosphere and in fluorescent lamps) or at very high temperatures (as in stars and nuclear fusion reactors).
- Plasma should be called 1st state of matter because it is what all the states arise from.



States of Matter



SOLID

Tightly packed, in a regular pattern
Vibrate, but do not move from place to place

LIQUID

Close together with no regular arrangement.
Vibrate, move about, and slide past each other

GAS

Well separated with no regular arrangement.
Vibrate and move freely at high speeds

PLASMA

Has no definite volume or shape and is composed of electrical charged particles

Occurrence of Plasma

Three forms of plasma

Plasmas occur naturally but can also be artificially made. Naturally occurring plasmas can be Earth-based (terrestrial) or space-based (astrophysical).

- There are three major types of Plasma i.e.

- Natural Plasma: Natural Plasma only exist at very high temperature or low temperature vacuum. It do not react rapidly but it is extremely hot (over 20,000 oC). There energy is so high that it vaporizes everything they touch.

- Artificial Plasma: Artificial Plasma can be created by ionization of a gas , as in neon signs. Plasma at low temperature is hard to maintain because outside a vacuum, low temperature plasma reacts rapidly with any molecule it encounters. This aspect makes this material, both very useful and hard to use.

- Terrestrial is a plasma layer that blankets the outer reaches of the Earth's atmosphere.

Plasma

Astrophysical plasma

All stars
Solar wind
Interstellar nebulae
Space between planets, star systems and galaxies

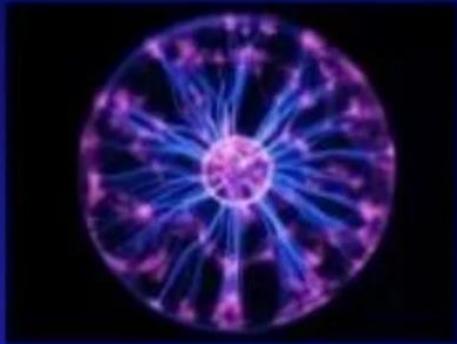
Terrestrial plasma

Lightning bolt
Auroras
Ionosphere
Extremely hot flames

Artificially produced

Plasma TVs
Fluorescent lighting
Plasma torch for cutting and welding
Plasma-assisted coatings

Natural Occurrence



Plasma can be found in sun and other stars ,lightning and the Aurora Borealis

Plasma

The Sun is an example of a star in its
plasma state



Plasma

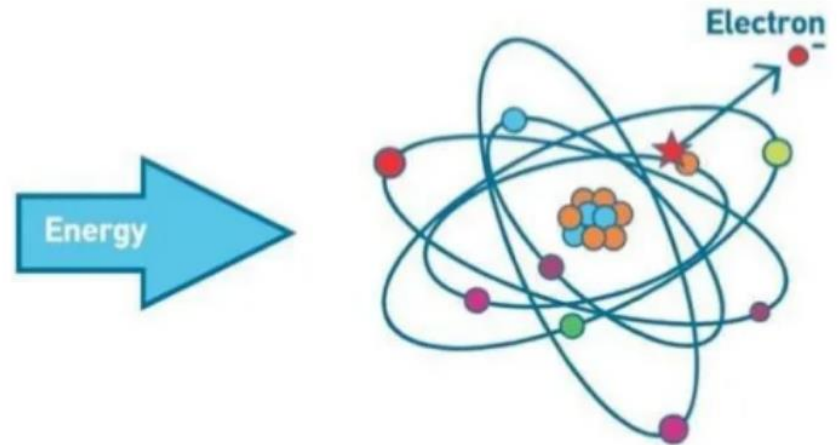
Some places where plasmas are found...

Extremely hot
Flames



Formation of Plasma

- When more heat is provided to atoms or molecules, they may be ionized. An electron may gain enough energy to escape its atom. After the escape of electron, atoms become ions. In sufficiently heated gas, ionization happens many times, creating clouds of free electrons and ions.
- This ionized gas mixture consisting of ions, electrons and neutral atoms is called PLASMA.



Plasma in Early Universe

- Over 99% of the matter in the visible universe is believed to be plasma. When the atoms in a gas are broken up, the pieces are called electrons and ions. Because they have an electric charge, they are pulled together or pushed apart by electric fields and magnetic fields. This makes a plasma act differently than a gas. For example, magnetic fields can be used to hold a plasma, but not to hold a gas. Plasma is a better conductor of electricity than copper.
- Plasma is usually very hot, because it takes very high temperatures to break the bonds between electrons and the nuclei of the atoms. Sometimes plasmas can have very high pressure, like in stars. Stars (including the Sun) are mostly made of plasma. Plasmas can also have very low pressure, like in outer space.

Classification of Plasma

Hot plasma (thermal plasma)

A hot plasma is one which approaches a state of local thermodynamic equilibrium (LTE). A hot plasma is also called a thermal plasma. Such plasmas can be produced by atmospheric arcs, sparks and flames.

Cold plasma (non-thermal plasma)

A cold plasma is one in which the thermal motion of the ions can be ignored. Consequently there is no pressure force, the magnetic force can be ignored, and only the electric force is considered to act on the particles. Examples of cold plasmas include the Earth's ionosphere (about 1000K compared to the Earth's ring current temperature of about 10^8K), the glow discharge in a fluorescent tube.

Classification of Plasma

Ultracold plasma

An ultracold plasma is one which occurs at temperatures as low as 1K. and may be formed by photoionizing laser-cooled atoms. Ultracold plasmas tend to be rather delicate, experiments being carried out in vacuum.

Plasma Ionization

The **degree of ionization** of a plasma is the proportion of charged particles to the total number of particles including neutrals and ions, and is defined as:

$$\alpha = n^+ / (n + n^+)$$

where n is the number of neutrals, and n^+ is the number of charged particles. α is the Greek letter alpha.

Plasma Frequency

The plasma frequency is the frequency above which the real part of the dielectric function becomes positive and the metal starts to behave like a dielectric. It varies with the electron number density n thus

$$\omega_p^2 = \frac{n e^2}{\epsilon_0 m}$$

where e is the charge on the electron, m is the effective mass of the electron, and ϵ_0 is the permittivity of free space.

Plasma period

$$\tau_p \equiv 1/\omega_p$$

Thermal Speed

Thermal velocity or thermal speed is a **typical velocity of the thermal motion of particles.**

Electron Thermal Velocity

$$v_{Te} = \left(\frac{kT_e}{m_e} \right)^{\frac{1}{2}}$$

Ion Thermal Velocity

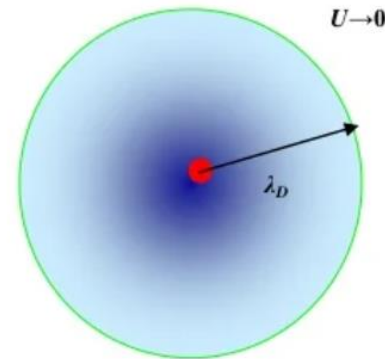
$$v_{Ti} = \left(\frac{kT_i}{m_i} \right)^{\frac{1}{2}}$$

Debye Length

The Debye length is a characteristic distance over which ions and electrons can be separated in a plasma (Chen et al., 1984) and is **equal to the ratio of the electron thermal velocity divided by the plasma frequency.**

$$\lambda_D \equiv \frac{\sqrt{T/m}}{\omega_p}$$

$$\lambda_D = \sqrt{\frac{\epsilon_0 T}{n e^2}}$$



- System size and time

$$\frac{\lambda_D}{L} \ll 1 \quad \frac{\tau_p}{\tau} \ll 1$$

- Debye shielding

Plasma Parameter

The plasma parameter is a dimensionless number, denoted by capital lambda Λ . The plasma parameter is usually interpreted to be the argument of the Coulomb logarithm, which is the ratio of the maximum impact parameter to the classical distance of closest approach in Coulomb scattering. In this case, the plasma parameter is given by

$$\Lambda = 4\pi n \lambda_D^3 = \frac{4\pi \epsilon_0^{3/2} T^{3/2}}{e^3 n^{1/2}}$$

where

n is the number density of electrons,

λ_D is the Debye length.

• Strong coupling $\Lambda \ll 1$

• Weak coupling $\Lambda \gg 1$

Basic Properties of Plasma

- Temperature
- Quasi-neutrality
- Thermal speed

$$T_s \equiv \frac{1}{3} m_s \langle v^2 \rangle$$

$$n_i \simeq n_e \equiv n$$

- Thermal speed

- $$v_{ts} \equiv \sqrt{2T/m_s} \quad v_{ti} \sim \sqrt{m_e/m_i} v_{te}$$

- Plasma frequency

$$\omega_p^2 = \frac{n e^2}{\epsilon_0 m}$$

- Plasma period

$$\tau_p \equiv 1/\omega_p$$

Production of Plasma

Natural

- Solar nebula
- planetary rings
- interstellar medium
- comet tails
- noctilucent clouds
- lightning

Man-made

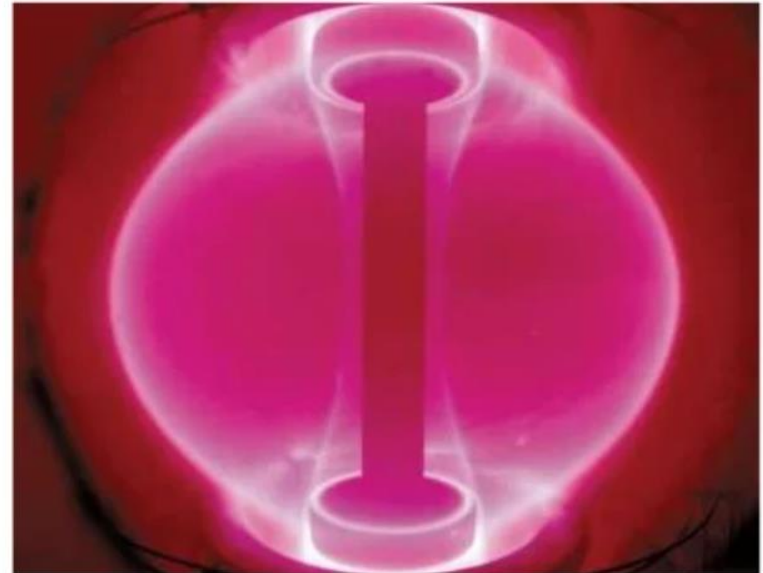
- Microelectronic processing
- rocket exhaust
- fusion devices

Application of Plasma

Fusion

Nuclear fusion is the process of recombining nuclei to form different nuclei and release vast amounts of energy. This is the process that powers the sun. If we can harness it, nuclear fusion has the potential to provide us with nearly limitless amounts of clean energy. As such, it is often described as the Holy Grail of plasma physics.

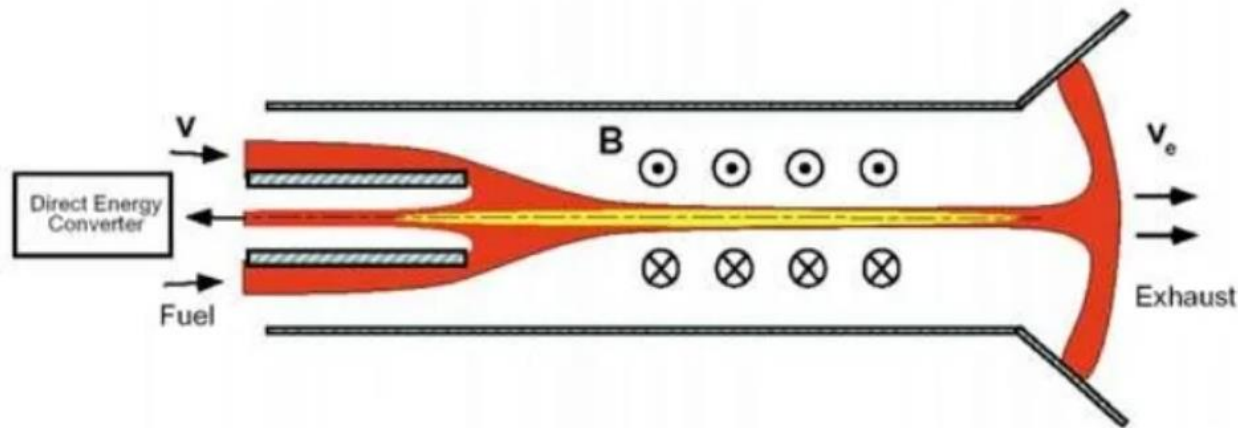
There are three conditions necessary for nuclear fusion: high temperatures (to about 10^7 K), high density, and prolonged stability. The high temperature requirement places us in the regime of plasmas. While experiments have attained these high temperatures, the primary difficulty is in achieving a sufficiently high combination of density and stability.



Application of Plasma

Propulsion in Space

Plasmas also have applications in the propulsion of spacecraft. The ZaP experiment is particularly well-suited to this application. Since it requires no externally applied magnetic field, the weight and size requirements of a such a vehicle are drastically lower than other plasma configurations would require. A diagram of a possible ZaP thruster design is shown in Figure



ZaP: Z-Pinch

Application of Plasma

Applications of Plasma

- Neon Signs are also made of Plasma.
- The glowing "gas" in a fluorescent bulb is Plasma.
- Plasma Globes are also used to make Plasma in Labs
- Plasma Televisions



Application of Plasma

Applications of Plasma

- Plasma drive lasers and particle accelerators.
- It's also used for making semi-conductors.
- Plasma lights up our homes, make our computers and electronic equipment

